

# AIPM: AI-based Process Management

-- Research Progress --

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# Outline

- ❑ AIPM objectives and approach
- ❑ Stakeholder needs
- ❑ System and functional requirements
- ❑ AIPM logical architecture
- ❑ Deriving AIPM systems architecture
- ❑ Current AIPM systems architecture
- ❑ R & D challenges
- ❑ Next steps

# AIPM: Core System Objectives

- ❑ Provide an **intelligent, model-based platform** for managing complex engineering and manufacturing processes end-to-end in the switchboard domain.
- ❑ Make **process knowledge explicit** by capturing tasks, dependencies, roles, resources, and changes in a form usable by both humans and AI.
- ❑ Support **planning, scheduling**, and **resource coordination** by linking process models to executable project-management information.
- ❑ Enable effective **change management** through traceability, version awareness, and impact analysis across design and manufacturing activities.
- ❑ Reduce **fragmented manual work** and reliance on tacit expert knowledge by integrating tools, data, and decision logic into one coherent environment.
- ❑ Improve **decision making, productivity**, and **quality** through AI-assisted reasoning, visibility of status and risks, and human-in-the-loop support.

# Our R & D Approach

- ❑ Build AIPM around a **formal process-modeling backbone** so workflows, dependencies, resources, and changes are explicitly represented rather than managed informally.
- ❑ Use an **agent-orchestrated architecture** in which specialized intelligent agents manage modeling, scheduling, version control, reasoning, and knowledge-retrieval functions.
- ❑ Integrate **specialty tools** such as **Visual Paradigm**, **PlantUML**, **Microsoft Project**, **GitLab**, and **GraphRAG** into a **unified operational toolchain** with defined roles and data flows.
- ❑ Combine **global and local LLM capabilities** with GraphRAG-based grounding so AI support is both powerful and **context-aware** while remaining adaptable to privacy and deployment needs.
- ❑ Develop the system incrementally through an **MVP-first strategy**, starting with core process modeling, scheduling, traceability, and AI-assisted decision support, then expanding capabilities over time.

MVP = Minimum Viable Product

# Stakeholder Needs

## ❑ Source of information

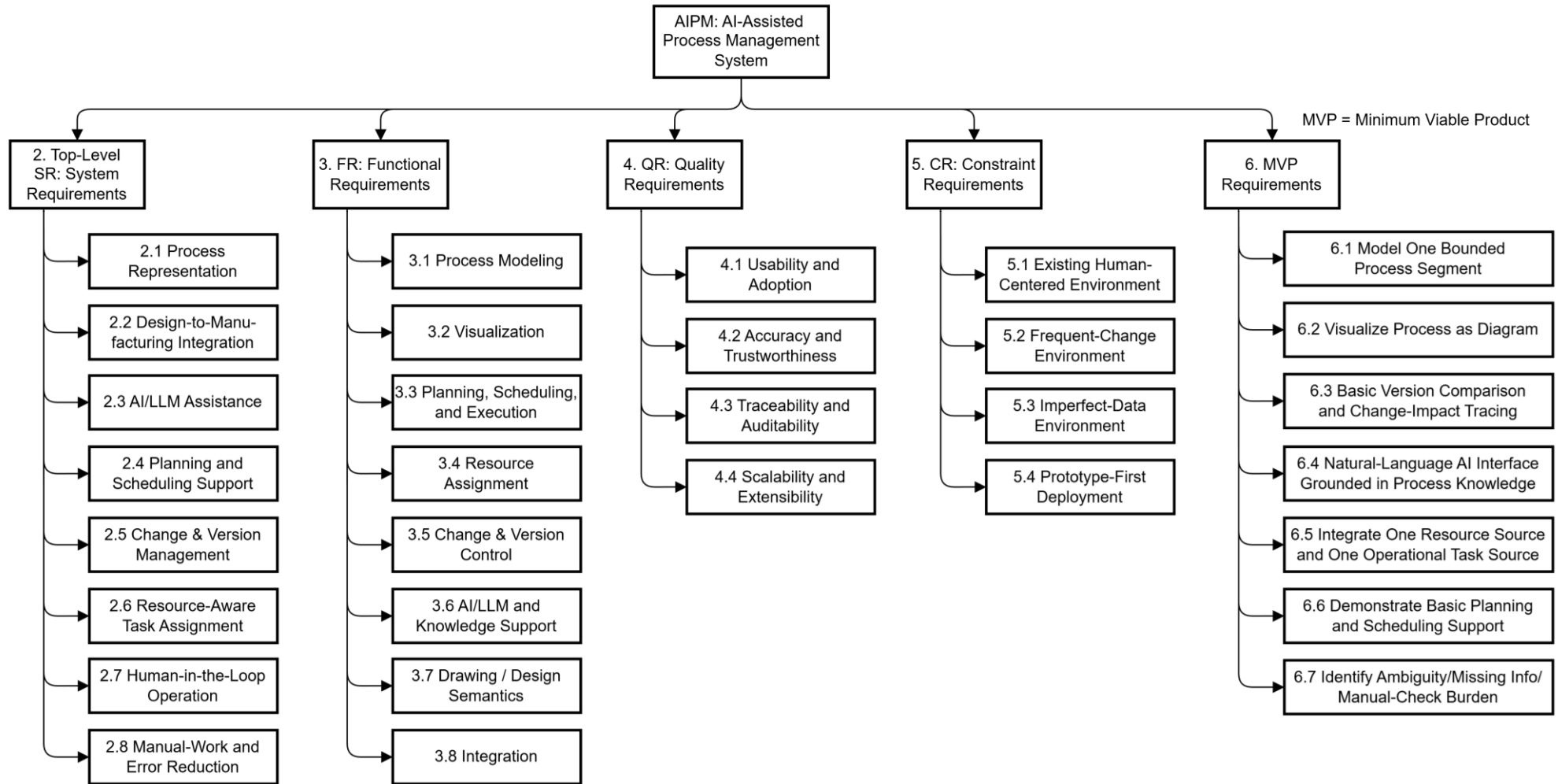
- Yasui San's email
- Process modeling experience

## ❑ Core stakeholder need:

- A digital process intelligence layer that can:
  - Represent the process explicitly,
  - Connect design and manufacturing,
  - Support planning and scheduling,
  - Visualize status and dependencies,
  - Manage change and versions,
  - Use enterprise data such as personnel/resource databases, and
  - Reduce manual work, overtime, outsourcing pressure, and human error.

- ▲ 2 Operational pain points driving these needs
  - 2.1 Overtime and outsourcing pressure
  - 2.2 Heavy reliance on tacit human knowledge
  - 2.3 Frequent changes requiring skilled judgment
  - 2.4 Fragmented current toolchain
- ▲ 3 Primary stakeholder needs
  - 3.1 Need to model the switchboard process explicitly in an MBSE-style form
  - 3.2 Need for diagrammatic visualization, not only back-end logic
  - 3.3 Need to connect AI/LLM capabilities directly into the process model
  - 3.4 Need for personnel/resource assignment integrated with company databases
  - 3.5 Need for planning and scheduling support similar to SEMAA mechanisms
  - 3.6 Need for visualization and version control for change management
- ▲ 4 Needs related to their digital transformation roadmap
  - 4.1 Need to align with BOP / MBOM / MES / PLM development
  - 4.2 Need to define and eliminate ambiguities before introducing BOM/MES
- ▲ 5 Needs around design digitization and semantic understanding
  - 5.1 Need to digitize not just drawings, but the meaning of drawings
  - 5.2 Need to reduce human checking burden and associated errors
  - 5.3 Need for an AI layer inside MBSE to unify tools and process knowledge
- ▲ 6 Context-specific requirements of the production environment
  - 6.1 Need to support high-mix, low-volume custom production
  - 6.2 Need to work in a human-centered production environment, not a fully automated one
- ▲ 7 Prototyping needs explicitly requested by the sponsors
  - 7.1 The sponsors want advice on where to start the MBSE prototype
  - 7.2 The sponsors want to know what information they should provide
- 8 Expansion and business value needs
- 9 A consolidated statement of stakeholder-needs
- 10 A summary list of stakeholder-needs
- 11 A prioritized list of stakeholder-needs

# AIPM: System & Functional Requirements



# Top-level System Requirements

## □ SR-1 Process representation

- The system shall provide a formal, machine-interpretable representation of the switchboard design and manufacturing process, including activities, sequencing, dependencies, inputs, outputs, roles, and resources.
- This follows directly from the request to create the switchboard process, generate a chart, and manage the process through MBSE.

## □ SR-2 Design-to-manufacturing integration

- The system shall represent and maintain relationships between design activities, manufacturing activities, and delivery-related process elements in a unified process model.
- This is required because the sponsors explicitly state that design is closely tied to manufacturing and want management across the full process.

# Top-level System Requirements

## □ SR-5 Change and version management

- The system shall support visualization and version control for process and design/manufacturing change management, including recording, comparison, and traceability of changes.
- This requirement comes directly from the sponsors' statement that planning/scheduling should also enable visualization and version control for change management.

## □ SR-6 Resource-aware task assignment

- The system shall support personnel and resource assignment by using enterprise/company database information about available resources and relevant attributes.
- This follows directly from the stated desire to handle personnel resource assignments by interacting with the company database.

# Top-level System Requirements

## □ SR-3 AI/LLM assistance

- The system shall provide an AI/LLM layer that can operate on the formal process model and linked project/manufacturing data to support interpretation, reasoning, querying, and coordination.
- This derives from the request to connect everything with LLM, to leverage GraphRAG and AI agent orchestration, and to understand the concept for process management.

## □ SR-4 Planning and scheduling support

- The system shall support planning and scheduling of process activities using model-based process information and resource information.
- This is explicitly requested through the question of whether planning and scheduling can be done using a SEMAA-like mechanism.

# Top-level System Requirements

## □ SR-7 Human-in-the-loop operation

- The system shall support human-in-the-loop decision making for cases involving ambiguity, incomplete information, or skilled judgment, rather than assuming full automation.
- This is required because the current environment relies heavily on human expertise and frequent change requires timely skilled judgments and adjustments.

## □ SR-8 Manual-work and error reduction

- The system shall reduce dependence on manual coordination, spreadsheet-driven task assignment, repeated human checking, and informal tacit-knowledge-based interpretation in the target process.
- This requirement is grounded in the sponsors' description of Excel-based task assignment, human interpretation of drawings, and human checks remaining a source of errors.



# Functional Requirements



## □ 3.1 Process modeling requirements

- **FR-1:** The system shall allow users to define and edit process models for switchboard design and manufacturing.
- **FR-2:** The system shall represent hierarchical process structures, including process, subprocess, task, and subtask levels.
- **FR-3:** The system shall represent precedence, dependency, and constraint relationships among tasks and process elements.
- **FR-4:** The system shall associate each process element with responsible roles, assigned personnel, required inputs, produced outputs, and related artifacts.
- **FR-5:** The system shall support process states such as planned, in progress, completed, blocked, changed, and awaiting review.
- **FR-6:** The system shall allow process elements to be linked to design documents, drawings, BOM/BOP/MBOM elements, manufacturing tasks, and change records.

# Functional Requirements



## □ 3.3 Planning, scheduling, and execution requirements

- **FR-11:** The system shall generate or support generation of project/process plans from the process model.
- **FR-12:** The system shall generate or support generation of schedules that consider dependencies, task durations, and resource constraints.
- **FR-13:** The system shall support rescheduling when process changes, design changes, or resource changes occur.
- **FR-14:** The system shall identify tasks at risk of delay due to change, dependency conflicts, or insufficient resource availability.
- **FR-15:** The system shall track execution progress against the plan and schedule.
- **FR-16:** The system shall support what-if analysis for alternative planning and scheduling scenarios.

# Functional Requirements



## □ 3.2 Visualization requirements

- **FR-7:** The system shall generate diagrammatic visualizations of the process model.
- **FR-8:** The system shall visualize dependencies, current status, bottlenecks, pending changes, and assigned resources.
- **FR-9:** The system shall provide visual comparison between process/model versions.
- **FR-10:** The system shall allow visualization filtered by lifecycle phase, department, role, product, process segment, or project state.

# Functional Requirements



## □ 3.4 Resource assignment requirements

- **FR-17:** The system shall retrieve personnel/resource information from the company database or connected enterprise data source.
- **FR-18:** The system shall assign or recommend personnel/resources for tasks based on availability and relevant matching criteria defined by the organization.
- **FR-19:** The system shall display current resource allocation and overload conditions.
- **FR-20:** The system shall support reassignment of tasks when changes or overload conditions arise.
- **FR-21:** The system shall preserve traceability between process tasks and assigned personnel/resources.



# Functional Requirements



## □ 3.5 Change and version-control requirements

- **FR-22:** The system shall maintain version histories for process models, process plans, schedules, and linked artifacts.
- **FR-23:** The system shall record who changed what, when, and why, for each managed artifact or process element.
- **FR-24:** The system shall detect and present the impacts of a design or process change on downstream manufacturing and scheduling elements.
- **FR-25:** The system shall support approval/review workflows for selected changes.
- **FR-26:** The system shall preserve traceability between a change request, affected process elements, impacted resources, and resulting schedule updates.

# Functional Requirements



## □ 3.7 Drawing/design semantics requirements

- **FR-33:** The system shall allow design artifacts, including drawings and symbols, to be linked to process and manufacturing elements.
- **FR-34:** The system shall support representation of the semantic meaning of selected drawing elements to the extent needed for process management and change analysis.
- **FR-35:** The system shall assist users in identifying where human review is required because design semantics are incomplete, ambiguous, or insufficiently digitized.
- **FR-36:** The system shall support impact tracing from design changes to manufacturing tasks and checks.



# Functional Requirements

## □ 3.6 AI/LLM and knowledge-support requirements

- **FR-27:** The system shall allow users to query the process and project/manufacturing knowledge base using natural language.
- **FR-28:** The system shall generate responses grounded in the linked process model and connected data sources rather than only using generic language generation.
- **FR-29:** The system shall support knowledge retrieval across structured and unstructured data relevant to process management.
- **FR-30:** The system shall identify missing, ambiguous, or inconsistent information in process descriptions or linked artifacts.
- **FR-31:** The system shall support explanation of recommendations, answers, or detected issues by referencing relevant process/model elements and source data.
- **FR-32:** The system shall support orchestration of multiple AI functions or agents for tasks such as retrieval, analysis, recommendation, and change-impact assessment.



# Functional Requirements

## □ 3.8 Integration requirements

- **FR-37:** The system shall integrate, or be designed to integrate, with relevant enterprise and operational systems such as company databases, Director6, Excel-originated task data, BOP, MBOM, MES, and PLM repositories.
- **FR-38:** The system shall support import/export or synchronization of process-related data with connected systems.
- **FR-39:** The system shall maintain referential consistency among linked external records and internal model elements.
- **FR-40:** The system shall tolerate partial or staged integration so that prototyping can begin before all source systems are fully formalized.

# Quality and performance requirements

## □ 4.1 Usability and adoption

- **QR-1:** The system shall provide interfaces understandable by engineering, manufacturing, and management stakeholders.
- **QR-2:** The system shall support incremental adoption without requiring the complete replacement of current tools at the outset.
- **QR-3:** The system shall make model and schedule information visible in a way that supports operational use, not merely documentation.

# Quality and performance requirements

## □ 4.3 Traceability and auditability

- **QR-7:** The system shall provide end-to-end traceability among stakeholder concerns, process elements, assignments, schedules, changes, and linked source artifacts.
- **QR-8:** The system shall preserve auditable histories of decisions, assignments, and model/schedule changes.



# Quality and performance requirements

## □ 4.2 Accuracy and trustworthiness

- **QR-4:** The system shall distinguish between confirmed information, inferred information, and uncertain information in AI-assisted outputs.
- **QR-5:** The system shall preserve links to source data or model elements used in AI-assisted recommendations and answers.
- **QR-6:** The system shall allow human review and override of AI-generated recommendations, assignments, or assessments.



# Quality and performance requirements

## □ 4.4 Scalability and extensibility

- **QR-9:** The system shall be architected so the initial prototype can be expanded to additional processes, products, or organizations.
- **QR-10:** The system shall support extension to broader marine equipment and shipyard process-management contexts.

# Constraint requirements

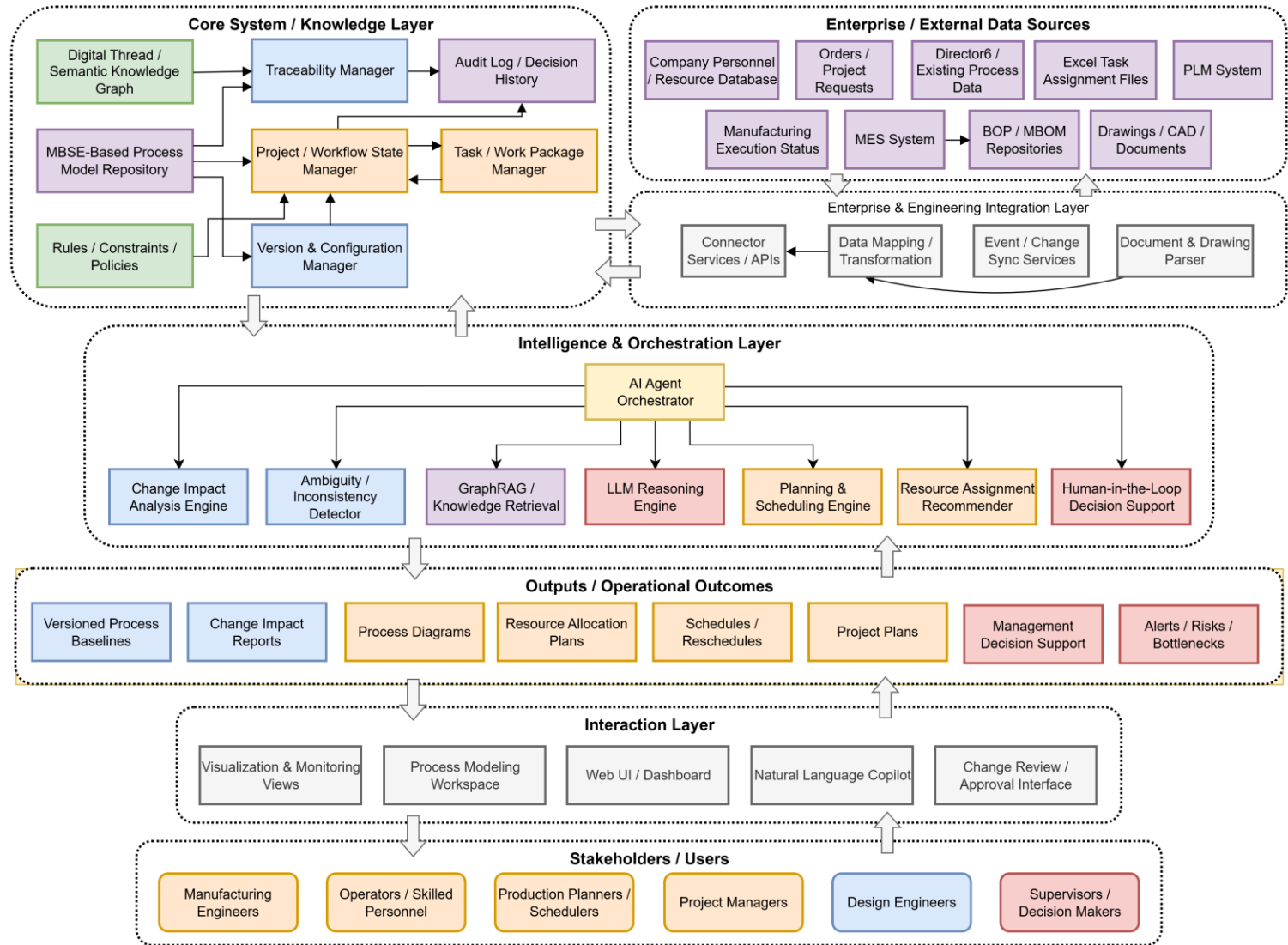
- ❑ **CR-1** Existing human-centered environment
  - The system shall operate effectively in an environment where many activities remain human-performed and not fully automated.
- ❑ **CR-2** Frequent-change environment
  - The system shall support high-mix, low-volume custom production with frequent changes during design and manufacturing.
- ❑ **CR-3** Imperfect-data environment
  - The system shall support operation under conditions where some process and design information remains ambiguous, incomplete, or under ongoing standardization.
- ❑ **CR-4** Prototype-first deployment
  - The system shall support phased prototyping, allowing an initial deployment to begin from either design or manufacturing scope and expand iteratively. This is based on the sponsors' explicit question about where to start prototyping and what information to provide.

# Requirements for an MVP

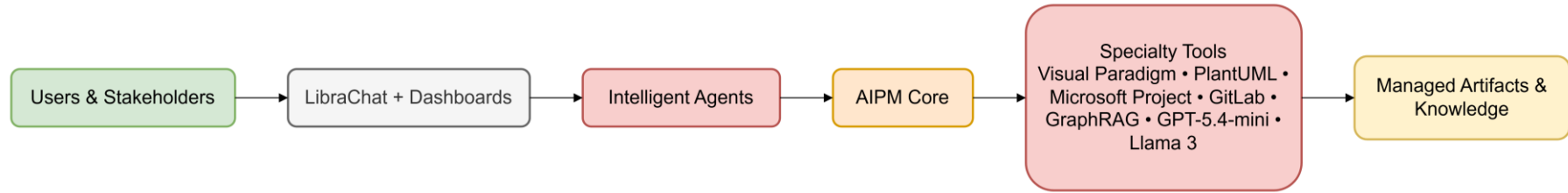
- ❑ **MVP-1**
  - The prototype shall model one bounded switchboard-related process segment with explicit tasks, dependencies, roles, and linked artifacts.
- ❑ **MVP-2**
  - The prototype shall visualize that process segment as a navigable diagram.
- ❑ **MVP-3**
  - The prototype shall support at least basic version comparison and change-impact tracing.
- ❑ **MVP-4**
  - The prototype shall support a natural-language AI interface grounded in the modeled process and selected source documents/data.
- ❑ **MVP-5**
  - The prototype shall integrate at least one personnel/resource data source and one operational task source.
- ❑ **MVP-6**
  - The prototype shall demonstrate planning/scheduling support for the selected process segment.
- ❑ **MVP-7**
  - The prototype shall identify at least one class of ambiguity, missing information, or manual-check burden for user review.

# AIPM System Logical Architecture

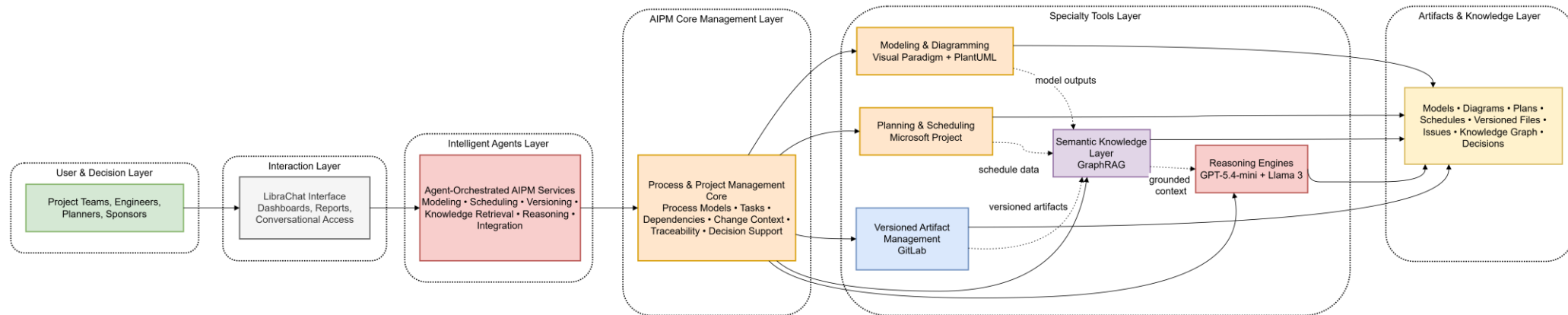
## Functional Color Code:



# Deriving AIPM System Architecture

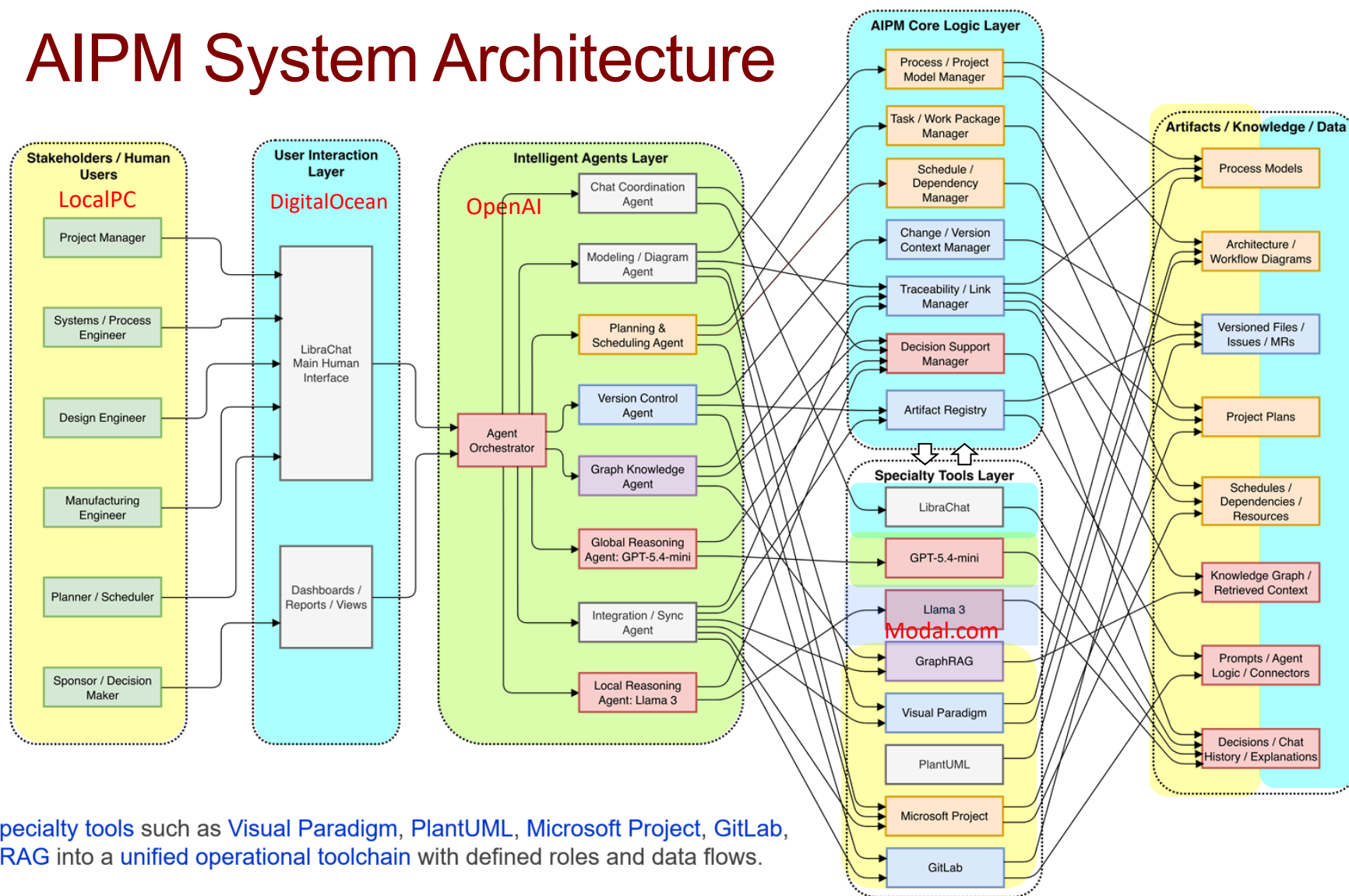


- Integrate **specialty tools** such as **Visual Paradigm**, **PlantUML**, **Microsoft Project**, **GitLab**, and **GraphRAG** into a **unified operational toolchain** with defined roles and data flows.





# AIPM System Architecture



- Integrate specialty tools such as Visual Paradigm, PlantUML, Microsoft Project, GitLab, and GraphRAG into a unified operational toolchain with defined roles and data flows.

# R & D Challenges

## ❑ System challenges:

- Building a **consistent semantic foundation** across **heterogeneous tools** and **data sources** so process models, schedules, diagrams, documents, versioned artifacts, and retrieved knowledge **remain aligned**.
- Managing **change propagation** across interconnected process elements, schedules, documents, decisions, and tool outputs without losing version integrity, consistency, or traceability.
- Preventing **error amplification** across **multiple agents and tools**, where one poor extraction, wrong retrieval, or mistaken interpretation could propagate through the larger AIPM workflow
- Evaluating and validating **system outputs** in domain-specific settings where correctness may be difficult to assess, especially for schedule recommendations, change-impact analysis, and interpretations of ambiguous engineering information.

## ❑ Knowledge challenges:

- Converting **informal, incomplete, multimodal, and expert-driven materials** such as notes, diagrams, charts, annotated visuals, emails, marked-up files, and partial documents into **usable datasets for RAG** and **possible fine-tuning** while preserving domain meaning & quality.

- Defining an **appropriate ontology/schema** for **processes**, tasks, roles, resources, artifacts, dependencies, and changes so all tools and agents interpret project information consistently.
- Maintaining **data quality, provenance, synchronization, and access control** across **evolving project repositories**, enterprise tools, and sensitive organizational information.

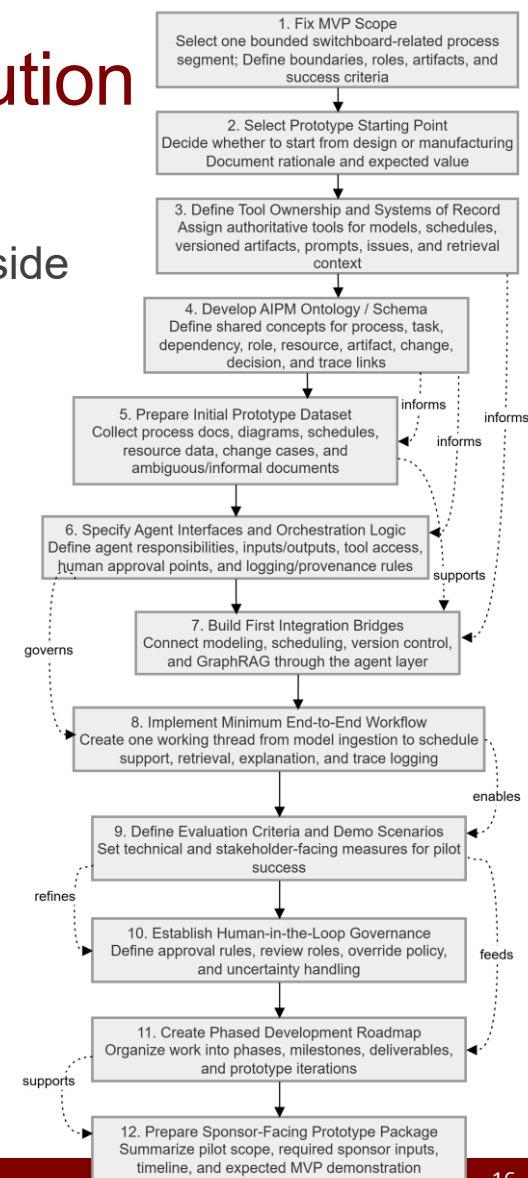
## ❑ AI & agentic technology related challenges:

- Enabling **reliable AI reasoning** and **agent behavior** that are grounded in accurate, traceable, and up-to-date project context rather than generic or hallucinated responses.
- Designing **effective retrieval pipelines**, including chunking, indexing, linking, graph construction, and ranking, so that RAG can provide **truly relevant context** for project and process management tasks.
- Determining the right balance among **prompt engineering, RAG, knowledge graphs**, and **fine-tuning** so the system gains domain capability without unnecessary complexity, cost, or maintenance burden.
- Supporting **human-in-the-loop decision making** for high-impact tasks such as **planning, scheduling, and change management**, while ensuring AI recommendations remain explainable and reviewable.



# Next Steps: Prototype Planning & Execution

1. Fix the scope of the **first MVP use case**
2. Select the prototype **starting point**: design side or manufacturing side
3. Define the **authoritative artifacts** and **tool ownership**
4. Develop the **AIPM ontology / schema**
5. Prepare the **initial prototype dataset**
6. Specify the **agent interfaces** and **orchestration logic**
7. Build the **first integration bridges**
8. Implement the **minimum end-to-end workflow**
9. Define **evaluation criteria** and **demonstration scenarios**
10. Establish **human-in-the-loop governance**
11. Create a **phased development roadmap**
12. Prepare the **user-facing prototype package**



# Thank You !

## Questions?